

Constructing Proof: Pythagoras, Trigonometry and Shape Properties

GeoGebra Geometry · GeoGebra

Years 8-10

~30 minutes

Mathematics · Geometry & Measurement

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.geogebra.org/geometry>

Curriculum links: AC9M8M06 · AC9M8SP02 · AC9M9M03 · AC9M10M03

Learning intentions

- I can use GeoGebra to construct and test the properties of triangles and quadrilaterals.
- I can apply Pythagoras' theorem and trigonometric ratios to solve for unknown lengths and angles.
- I can explain which geometric properties stay the same when a shape is transformed.

Getting started

Open GeoGebra Geometry and use the Polygon tool to build shapes on the grid. Remember that when you drag a vertex, GeoGebra automatically keeps any angles, side lengths or parallel relationships you've set up — this lets you test whether a property is a coincidence or a rule. Keep your constructions on screen as you work through the tasks below.

Tasks

1. Construct a right-angled triangle with the Polygon tool. Use the Distance or Length tool to measure the two shorter sides (legs) and the hypotenuse. Record all three lengths.

2. Using your measured leg lengths, calculate $a^2 + b^2$ and compare it with c^2 (the hypotenuse squared). How close are the two values?

Working:

3. Drag one vertex of your triangle to change its shape and size, keeping it right-angled. Remeasure the three sides and check whether $a^2 + b^2 = c^2$ still holds.

4. Construct a quadrilateral using two congruent triangles (for example, join two triangles along a shared side to make a parallelogram). Drag a vertex and describe which angles or side lengths stay equal, and which change.

5. Construct three different right-angled triangles of different sizes. For each, record the two legs and the hypotenuse in the table below.

Trial	Leg a (m)	Leg b (m)	Hypotenuse c (m)

6. Construct a right-angled triangle to represent a ladder leaning against a wall: a vertical side for the wall, a horizontal side for the ground, and a hypotenuse for the ladder. Use the Angle tool to set the angle between the ladder and the ground to 60° . If the ladder is 4 m long, calculate the height it reaches up the wall using trigonometry, then check your answer against the construction.

Working:

7. A surveyor stands 30 m from the base of a tower and measures the angle of elevation to the top as 40° . Construct this situation in GeoGebra as a right-angled triangle. Use your construction and trigonometry to estimate the height of the tower, and explain your method.

8. Construct two similar triangles by scaling one triangle from a fixed point (redraw a larger copy keeping the same angles). Measure corresponding sides on both triangles and calculate the ratio between them.

Extension (optional)

Choose a real object near you (a tree, flagpole, or building) that you could measure using the 'angle of elevation' method from Task 7. Describe what two measurements you would need to take in real life, and how you'd use trigonometry to calculate its height.
